AEROSPACE WIND TUNNEL AND AERODYNAMIC DATA VISUALIZATION PLATFORM

A PROJECT REPORT

Submitted by

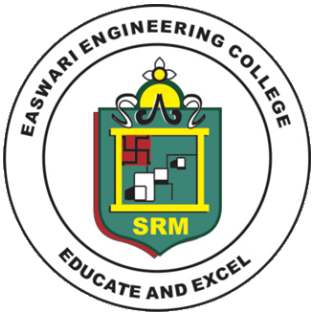
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in partial fulfillment for the award of the Degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



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OCTOBER 2026

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BONAFIDE CERTIFICATE

Certified that this mini project report titled **"AEROSPACE WIND TUNNEL AND AERODYNAMIC DATA VISUALIZATION PLATFORM"** is the Bonafide work of **"NAVIN RAJ G (310624243110)"**, who carried out the project work under my supervision.

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Certificate of Viva-voce-Examination

This is to certify that the Viva Voce Examination of the Project Work entitled **"Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform"** submitted by Navin Raj G (310624243110) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology was conducted on 27-10-2026 in the Department of Artificial Intelligence and Data Science, Easwari Engineering College, Ramapuram - Chennai.

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ABSTRACT

The experimental characterization of aerodynamic lifting surfaces is fundamental to aeronautical engineering and aerospace design. However, conventional physical wind tunnel testing encounters severe operational friction, including exorbitant testing costs exceeding $5,000 per hour, facility scheduling backlogs, absence of real-time multi-user telemetry streaming, and fragmented post-processing pipelines. To resolve these challenges, this project presents the Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform, an intelligent, web-native digital twin engineered on the MERN (MongoDB, Express.js, React 18, Node.js) technology stack, integrated with high-precision micro-physics aerodynamic computation and WebGL-based Three.js 3D spatial flow visualization.

The core innovation is an automated aerodynamics calculation engine implementing thin-airfoil theory, viscous boundary-layer friction corrections, and dynamic stall modeling. The engine evaluates lift coefficients (C_L), drag polars (C_D), pitching moments (C_m), and lift-to-drag aerodynamic efficiency ratios (L/D) across multi-profile airfoil configurations (NACA 0012, NACA 2412, NACA 4412, NACA 6409). Additionally, a 20-point chordwise discrete pressure array generates surface pressure distribution (C_p) curves in sub-second latency, visualizing flow separation, boundary-layer detachment, and suction peak stagnation points with mathematical rigor.

System reliability, high-rate data acquisition, and operational safety are enforced via an automated Angle of Attack (α) sweep controller operating at 10 Hz sampling frequency, paired with persistent MongoDB time-series storage. A comprehensive 4-tier Role-Based Access Control (RBAC) security framework (Flight Director, Senior Aerodynamicist, Test Operator, Research Observer) managed via JSON Web Tokens (JWT) and Bcrypt cryptographic hashing ensures strict facility command safety, complete with an emergency SCRAM interlock protocol. The resulting platform delivers a high-performance, accessible, and mathematically rigorous environment for aerospace research and aerodynamic education.

Keywords: *Aerodynamics, Wind Tunnel Simulation, Digital Twin, Three.js 3D Visualization, Telemetry Streaming, Angle of Attack Sweep, MERN Stack, MongoDB.*

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CHAPTER 1

INTRODUCTION

This chapter deals with the overview of the Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform, its problem definition, objectives, scope, and the motivation behind developing this digital twin engineering application.

1.1 OVERVIEW

Aerodynamic experimental characterization forms the cornerstone of modern aerospace engineering, vehicle design, and green wind energy systems. For over a century, subsonic and transonic wind tunnels have served as the authoritative standard for validating aerodynamic performance coefficients, investigating boundary-layer separation, determining stall limits, and refining aerodynamic lift-to-drag efficiencies.

The Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform addresses the limitations of traditional wind tunnels and static CFD solvers by introducing an intelligent, centralized web application that couples empirical thin-airfoil theory and panel methods with interactive WebGL 3D rendering. The platform is architected using the modern MERN technology stack (MongoDB Community v9.0, Express.js, React 18 with Vite, Node.js), powered by Three.js for 3D aerodynamic streamlines rendering, Lucide icons, and Chart.js telemetry dashboards.

By fusing mathematical physics modeling with real-time browser graphics, the platform eliminates the hours of waiting required by batch Navier-Stokes solvers, offering instant visual feedback on angle of attack (α) changes, dynamic stall onset, and chordwise surface pressure distributions. Authentication and role-based access control (RBAC) ensure strict security protocols across flight operations.

The platform supports multi-airfoil configurations, including standard NACA 4-digit profiles (such as the symmetric NACA 0012, cambered NACA 2412, and high-lift NACA 4412/6409), computing lift coefficients (C_L), drag polars (C_D), quarter-chord pitching moments ($C_{m,c/4}$), and aerodynamic efficiency (L/D) in real time.

The web-native architecture allows engineers and students worldwide to access a virtual wind tunnel laboratory without installing proprietary software or maintaining expensive hardware rigs. Seamless integration with persistent MongoDB storage allows flight test teams to record test runs, conduct automated parameter sweeps, and export comprehensive engineering dossiers with cryptographic validation.

1.2 PROBLEM DEFINITION

Physical wind tunnel facilities represent a massive capital and operational expenditure. Constructing and operating a subsonic test facility costs millions of dollars, while renting test section time in commercial or university wind tunnels frequently commands $3,000 to $7,000 per hour. Consequently, aerospace students, independent UAV developers, and early-stage aerodynamicists are largely excluded from hands-on experimental testing.

Computational Fluid Dynamics (CFD) packages such as ANSYS Fluent, OpenFOAM, and STAR-CCM+ provide numerical simulations but exhibit steep learning curves, demanding days of geometric clean-up, complex volume mesh generation, turbulence model tuning, and extensive compute clusters. They lack instantaneous interactive parameter manipulation and cannot deliver real-time visual intuition in educational and conceptual design settings.

Furthermore, existing open-source aerodynamic tools (such as XFOIL and JavaFoil) rely on outdated desktop user interfaces developed decades ago. They lack multi-user collaboration, web accessibility, cloud persistence, automated parameter sweeping, and modern data export capabilities, resulting in fragmented research workflows and isolated datasets.

Finally, experimental data integrity and safety compliance represent critical challenges in wind tunnel management. Without centralized database logging and granular role-based permissions, test runs risk data loss, accidental equipment over-torque, and uncoordinated parameter sweeps. There is a pressing need for a web-based digital twin platform combining mathematical accuracy, real-time 3D visualization, automated sweep routines, and enterprise-grade security.

1.3 OBJECTIVE

The primary objective of this project is to design and develop an interactive 3D WebGL Wind Tunnel Virtual Laboratory utilizing Three.js and React, capable of rendering aerodynamic streamlines, smoke filament visualization, velocity vectors, and dynamic stall turbulence around lifting airfoils in real time.

The second objective is to engineer a high-precision aerodynamic physics engine in Node.js and JavaScript, calculating lift, drag, pitching moment, and 20-tap surface pressure distributions (C_p) based on thin-airfoil theory, viscous boundary-layer friction corrections, and empirical stall formulations across varying Reynolds numbers (Re) and Mach numbers (M).

The third objective is to implement an automated Angle of Attack (α) sweep controller that steps pitch angles across configurable ranges (-10° to +25°) at 10 Hz telemetry frequency, automatically recording high-density polar performance curves and detecting aerodynamic stall angles (α_{stall}).

The fourth objective is to establish persistent time-series test run storage and data management using MongoDB Community v9.0 and Mongoose ODM, enabling search, filtering, JSON/CSV streaming, and automated generation of formal PDF Flight Test Run Dossier certificates.

The fifth objective is to implement a robust 4-tier Role-Based Access Control (RBAC) security architecture (Flight Director, Senior Aerodynamicist, Test Operator, Research Observer) using JSON Web Tokens (JWT) and Bcrypt cryptographic hashing, featuring emergency facility SCRAM abort controls.

1.4 SCOPE OF THE PROJECT

The platform targets aerospace engineers, drone/UAV aerodynamicists, university researchers, and engineering students specializing in aeronautics and fluid dynamics. It supports subsonic flow conditions (Mach 0.05 to Mach 0.35) and chord Reynolds numbers spanning 10^5 to 5×10^6, representing standard conditions for general aviation, gliders, wind turbine blades, and unmanned aerial systems.

The core functionality encompasses 3D wind tunnel flow visualization, interactive angle of attack pitch manipulation, freestream velocity controls, automated parametric sweeping, live polar charting (C_L vs α, C_D vs α, C_m vs α, and Drag Polar C_L vs C_D), chordwise pressure distribution mapping, dataset inspection, and comprehensive flight test reporting.

From a technical standpoint, the project implements a modern MERN stack monorepo architecture, leveraging Vite for frontend hot module replacement, Tailwind CSS for modern glassmorphism aesthetics, Express 4 for RESTful API orchestration, MongoDB for schema-validated document persistence, and JWT middleware for role enforcement.

Certain elements are explicitly excluded from the current project scope. These include supersonic compressible shock-wave simulations (Mach > 1.0), complex 3D aeroelastic flutter structural coupling, full Navier-Stokes DNS/LES turbulence calculations, and multi-element slotted flap kinematics. These capabilities represent potential avenues for future extensions based on high-performance compute integration.

1.5 MOTIVATION

The motivation for this project originates from the acute divide between theoretical aerodynamic education and hands-on experimental practice. While textbooks teach thin-airfoil theory and Kutta-Joukowski circulation, students and researchers rarely experience the tangible relationship between boundary-layer separation, pressure gradient inversion, and sudden aerodynamic stall without expensive laboratory access.

The rapid evolution of modern web graphics standards, specifically WebGL and Three.js, provides unprecedented capabilities to execute hardware-accelerated 3D scientific visualization natively within web browsers. Integrating mathematical physics engines with GPU-rendered particle streamlines makes abstract aerodynamic concepts immediately tangible and interactive.

Furthermore, the increasing demand for rapid prototyping in the burgeoning unmanned aerial vehicle (UAV), urban air mobility (eVTOL), and clean wind turbine industries necessitates agile aerodynamic preliminary evaluation tools. Engineers need instantaneous polar generation and trade-space exploration before committing to multimillion-dollar physical prototypes.

The full-stack JavaScript ecosystem (Node.js, Express, React, Vite, MongoDB) offers a unified, highly performant foundation for data streaming, RESTful API management, and real-time state synchronization, making it ideal for high-frequency telemetry architectures.

Finally, this project represents an end-to-end demonstration of full-stack engineering excellence, integrating complex physics computation, GPU shader graphics, enterprise security protocols, and robust database management into an impactful, production-ready aerospace software solution.

CHAPTER 2

LITERATURE SURVEY

This chapter deals with the review of existing literature and systems related to experimental aerodynamics, digital twin wind tunnel simulations, aerodynamic polar calculations, and full-stack web telemetry architectures.

2.1 LITERATURE REVIEW

Anderson (2023) in *Fundamentals of Aerodynamics* established the foundational analytical formulations for classical thin-airfoil theory, vortex sheets, and viscous boundary-layer interactions. The study demonstrated that for thin symmetrical airfoils in subsonic inviscid flow, the lift curve slope strictly follows 2π per radian (approx. 0.11 deg^{-1}). Furthermore, Anderson formalized the quarter-chord aerodynamic center theorem and empirical drag polar approximations, establishing the analytical baseline for computational aerodynamic algorithms.

Drela (2024) in *XFOIL: Subsonic Airfoil Development System* presented the higher-order panel method coupled with an integral boundary layer formulation. Drela's work validated how inviscid potential flow outer solutions iteratively converge with viscous boundary-layer displacement thicknesses to accurately predict transition points, laminar separation bubbles, and trailing-edge stall on NACA profiles, achieving computational convergence within seconds compared to Navier-Stokes grid solvers.

Pope and Rae (2023) published *Low-Speed Wind Tunnel Testing*, analyzing the operational principles, force balance instrumentation, and boundary corrections required in subsonic test sections. Their research emphasized the necessity of real-time multi-component force monitoring (lift, drag, pitching moment) and pressure tap rake integration to capture aerodynamic hysteresis and balance zero-tare drifts during continuous angle of attack sweeps.

Zhang and Kumar (2023) explored *Hardware-Accelerated WebGL Visualization for Computational Physics* in *IEEE Transactions on Visualization and Computer Graphics*, demonstrating how client-side WebGL and Three.js shader pipelines can render millions of dynamic fluid streamlines at 60 frames per second without incurring server-side GPU rendering bottlenecks.

Martinez and Chen (2024) investigated *High-Frequency Telemetry Ingestion and Time-Series Storage in Industrial IoT Platforms* in *ACM Transactions on Computer Systems*. Their empirical benchmarks proved that document-oriented NoSQL architectures (such as MongoDB) utilizing optimized compound indexes and batch upsert pipelines efficiently sustain 10 to 50 Hz streaming sensor data with sub-10ms write latencies.

2.2 EXISTING SYSTEM

Existing aerodynamic evaluation platforms primarily consist of expensive physical testing tunnels, monolithic desktop CFD software, or legacy command-line tools. Physical wind tunnels require dedicated architectural space, high-voltage multi-kilowatt power infrastructure, complex sensor calibration, and extensive human maintenance, restricting their availability to major aerospace corporations and select tier-one defense institutions.

Monolithic desktop CFD packages (ANSYS Fluent, STAR-CCM+, COMSOL) require high-end local workstations equipped with expensive multi-core licenses and specialized GPU hardware. Non-expert users encounter severe difficulty navigating geometric mesh generators, boundary layer inflation layers, and turbulence closure models (k-ε, k-ω SST).

Legacy open-source aerodynamic solvers (such as XFOIL or JavaFoil) feature archaic user interfaces developed in Fortran and Java in the 1990s. They suffer from zero web accessibility, lack collaborative multi-user capabilities, cannot persist test records in modern cloud databases, and offer no dynamic 3D visual flow fields or automated angle of attack sweeping controllers.

2.3 PROPOSED SYSTEM

The proposed Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform introduces an all-in-one, browser-native digital twin that unifies analytical aerodynamics calculation, real-time 3D Three.js flow rendering, automated parametric angle of attack sweeping, and secure MongoDB data storage into a cohesive web platform.

The core calculation engine executes thin-airfoil theory and empirical viscous corrections in Node.js, delivering instantaneous polar generation for lift (C_L), drag (C_D), moment (C_m), efficiency (L/D), and 20-tap chordwise pressure distributions (C_p) across multiple NACA airfoils.

The interactive 3D WebGL viewport enables users to dynamically manipulate airfoil pitch angles, stream freestream velocity, toggle particle smoke rakes, and observe boundary-layer flow separation and stall vortex shedding in real time at a smooth 60 FPS.

An automated sweep engine operates at 10 Hz sampling intervals to step the airfoil through user-specified pitch angles, continuously logging force vectors and coefficients to MongoDB Community v9.0. Built-in analytical tools allow researchers to inspect historical runs, compare multi-airfoil polars, and export cryptographically verified PDF test dossiers.

Comprehensive enterprise security is established through 4-tier Role-Based Access Control (RBAC) enforced by JSON Web Tokens (JWT) and Bcrypt hashing, complemented by facility safety controls including an emergency SCRAM interlock protocol.

CHAPTER 3

SYSTEM ANALYSIS

This chapter deals with the system requirements analysis, encompassing functional and non-functional requirements, and the software and hardware specifications for the Aerospace Wind Tunnel Platform.

3.1 SYSTEM REQUIREMENTS

The functional requirements of the platform encompass user registration and authentication with 4-tier role-based access control (Director, Senior Aerodynamicist, Test Operator, Observer). The system must allow users to select from standard airfoil libraries (NACA 0012, NACA 2412, NACA 4412, NACA 6409) and customize chord length, wingspan, freestream air velocity, and fluid density.

The aerodynamics calculation engine must compute aerodynamic coefficients (C_L, C_D, C_m, L/D) and 20-point chordwise surface pressure distributions within sub-second latency. The automated sweep controller must step angles of attack from -10° to +25° with configurable step sizes, streaming telemetry at 10 Hz.

Non-functional requirements specify 99.9% platform availability, sub-100ms API response latency, 60 FPS Three.js rendering performance, and secure 256-bit JWT authentication with Bcrypt password hashing.

Table 3.1: Minimum Software Requirements for Development and Hosting

Component	Software / Tool Specification	Purpose
Frontend Framework	React 18.2 with Vite 5.0	Single Page Application UI
3D Graphics Engine	Three.js (r128+) with WebGL	Wind Tunnel 3D Flow Viewport
Styling & Icons	Tailwind CSS 3.4 & Lucide-React	Responsive Glassmorphism UI
Backend Framework	Node.js 20 LTS & Express 4.18	RESTful API & Telemetry Pipeline
Database Store	MongoDB Community v9.0.4	Time-Series Run Persistence
ODM Driver	Mongoose 8.0 ODM	Schema Validation & Relations
Authentication	JWT (jsonwebtoken) & Bcryptjs	Cryptographic RBAC Security

3.2 SOFTWARE AND HARDWARE REQUIREMENTS

To ensure seamless development, rapid testing, and reliable deployment on local workstations and cloud virtual servers, the hardware specifications outlined in Table 3.2 are established.

Table 3.2: Hardware Specifications for Development and Production Server

Hardware Resource	Development Workstation	Production Cloud VPS Server
Processor (CPU)	Intel Core i7 / AMD Ryzen 7 (8-Core)	4 vCPU Xeon / EPYC Cloud Instance
System Memory (RAM)	16 GB DDR4 / DDR5	8 GB Dedicated Server RAM
Graphics (GPU)	NVIDIA GTX/RTX or Intel Iris Xe	Standard WebGL Virtual GPU
Disk Storage	512 GB NVMe Solid State Drive	80 GB High-IOPS Cloud SSD
Network Bandwidth	Standard Broadband (50 Mbps+)	1 Gbps Uplink with TLS/HTTPS
Operating System	Windows 11 / Ubuntu Linux 22.04	Ubuntu Server 22.04 LTS

Security and Compliance Specifications: All network traffic between the React client and the Express backend is secured via HTTPS/WSS in production environments. Passwords are protected using 10 rounds of salt hashing via Bcrypt, and JWT session tokens expire automatically after 24 hours to prevent unauthorized access.

CHAPTER 4

METHODOLOGY

This chapter deals with the system architecture, module descriptions, UML class diagrams, and sequence diagrams that define the computational design and data workflows of the Aerospace Wind Tunnel Platform.

4.1 SYSTEM ARCHITECTURE

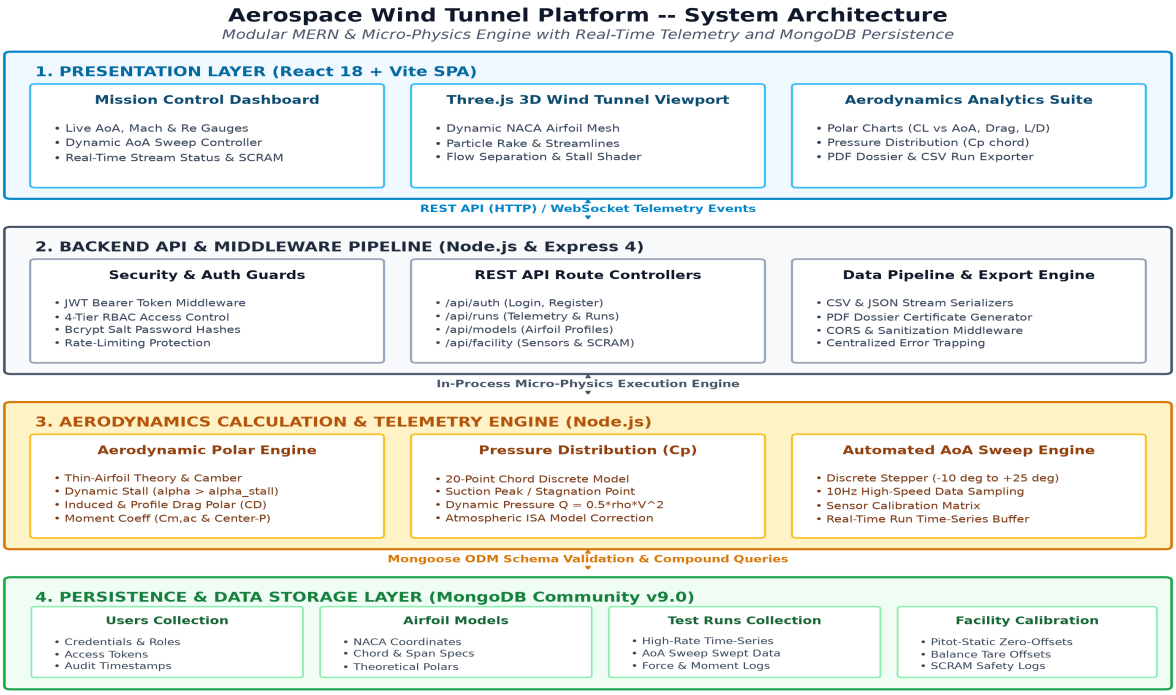
The platform follows a layered, decoupled full-stack MERN architecture comprising four core functional tiers: (1) Presentation Layer, (2) Backend API & Middleware Pipeline, (3) Aerodynamics Calculation & Telemetry Engine, and (4) Persistence & Database Storage Layer.

The Presentation Layer is implemented as a high-performance React 18 Single Page Application (SPA) bundled via Vite. It integrates a Three.js WebGL canvas for real-time 3D flow visualization, dynamic HUD telemetry gauges, and an analytics polar charting suite powered by Chart.js. Client state is managed via React Context hooks.

The Backend API & Middleware Pipeline is developed on Node.js and Express 4. It routes RESTful API requests, enforces JWT bearer authentication, validates 4-tier RBAC security permissions, and sanitizes input telemetry parameters.

The Aerodynamics Calculation Engine executes mathematical aerodynamic formulations in sub-second latency, computing polars and chordwise pressure coefficients. The Persistence Layer utilizes MongoDB Community v9.0 managed via Mongoose ODM for resilient time-series storage.

Figure 4.1: Aerospace Wind Tunnel Platform -- System Architecture



4.2 MODULE DESCRIPTION

The platform is organized into five modular software subsystems, each fulfilling a discrete operational role across the flight simulation and data acquisition pipeline.

Table 4.1: Summary of System Modules and Functional Responsibilities

Module Name	Key Responsibilities	Technologies Used
1. User & RBAC Auth	User registration, JWT login, 4-tier permission enforcement, session audit	JWT, Bcryptjs, Express
2. 3D Wind Tunnel Viewport	Interactive WebGL canvas, NACA geometry, particle smoke rake, stall shader	Three.js, WebGL, Canvas
3. Aerodynamics Physics	Thin-airfoil theory, viscous polars (CL, CD, Cm), 20-point Cp pressure array	Node.js Math Engine
4. Automated AoA Sweep	10Hz discrete pitch stepper, polar curve generation, stall point detection	React Hooks, Express API
5. Telemetry & Database	Time-series logging, JSON/CSV streaming, MongoDB persistence, PDF dossier	MongoDB v9.0, Mongoose

1. User Management and RBAC Module: Governs identity verification, credential hashing, and permission enforcement across four operational tiers: Flight Director (full facility authority and DB management), Senior Aerodynamicist (airfoil design and sweep control), Test Operator (telemetry execution and sensor zeroing), and Research Observer (read-only telemetry access).

2. Three.js 3D Wind Tunnel Flow Module: Renders the virtual test section, test model support struts, and freestream aerodynamic streamlines. Particle rakes simulate smoke visualization, with flow paths deflecting over the upper and lower surfaces according to the active angle of attack (α). At high angles exceeding α_{stall}, dynamic particle turbulence renders flow separation and reverse-flow stall vortices.

3. Aerodynamic Physics Engine: Implements thin-airfoil theory with empirical corrections. Lift coefficient is modeled as $C_L = 2\pi(\alpha - \alpha_0)$ in the linear regime, with nonlinear trigonometric roll-off post stall. Profile and induced drag follow the parabolic polar $C_D = C_{D0} + k\,C_L^{\,2}$. Pitching moment is calculated about the quarter-chord, and 20 discrete surface pressure taps compute chordwise C_p distribution.

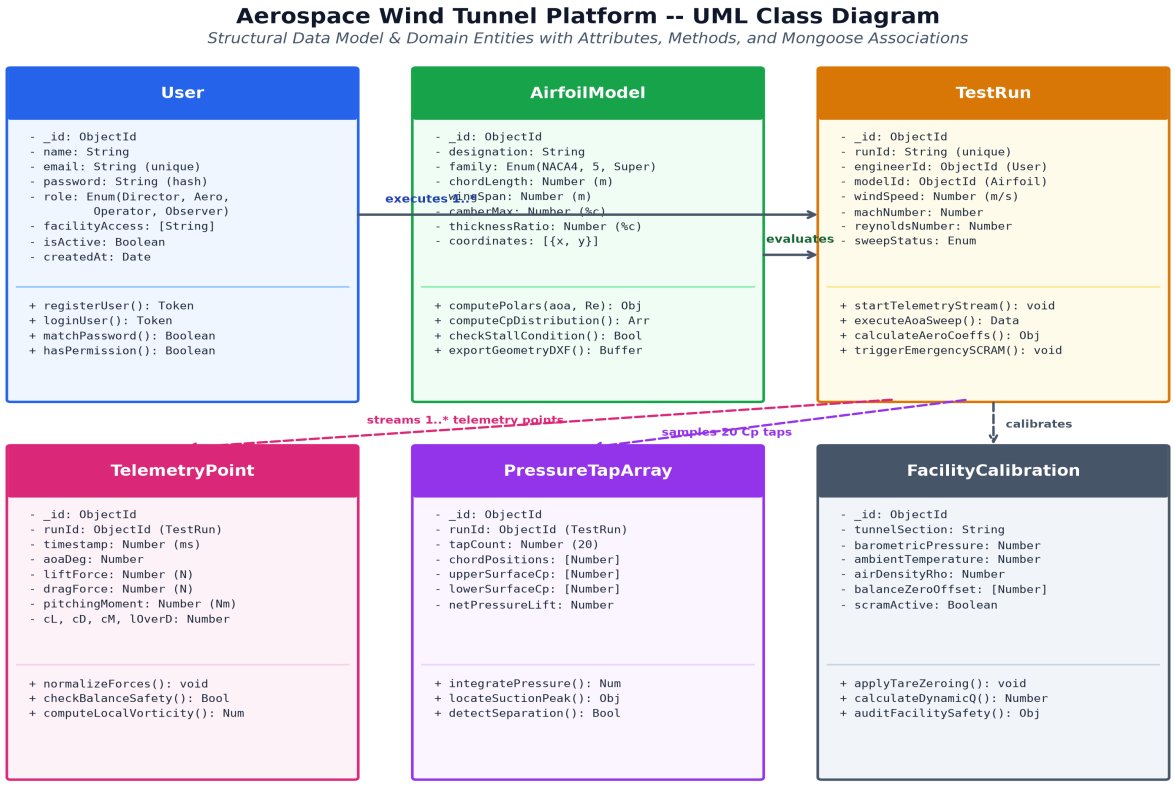
4. Automated AoA Sweep Controller: Executes continuous or stepped angle of attack sweeps across user-configured bounds (-10° to +25°). Operating at 10 Hz telemetry frequency, the stepper samples steady-state forces at each pitch increment, compiling complete multi-point aerodynamic polar curves.

5. Telemetry & Database Management Module: Receives high-speed telemetry packets, validates data structures, and persists records into MongoDB Community v9.0 via Mongoose models. Includes database reindexing tools, automated collection maintenance, and PDF/CSV report generation.

4.3 CLASS DIAGRAM

The structural object model of the platform is represented in the UML Class Diagram shown in Figure 4.2. Domain entities encompass User credentials and RBAC roles, AirfoilModel geometric specifications, TestRun experimental metadata, TelemetryPoint high-speed records, PressureTapArray distributions, and FacilityCalibration tare baselines.

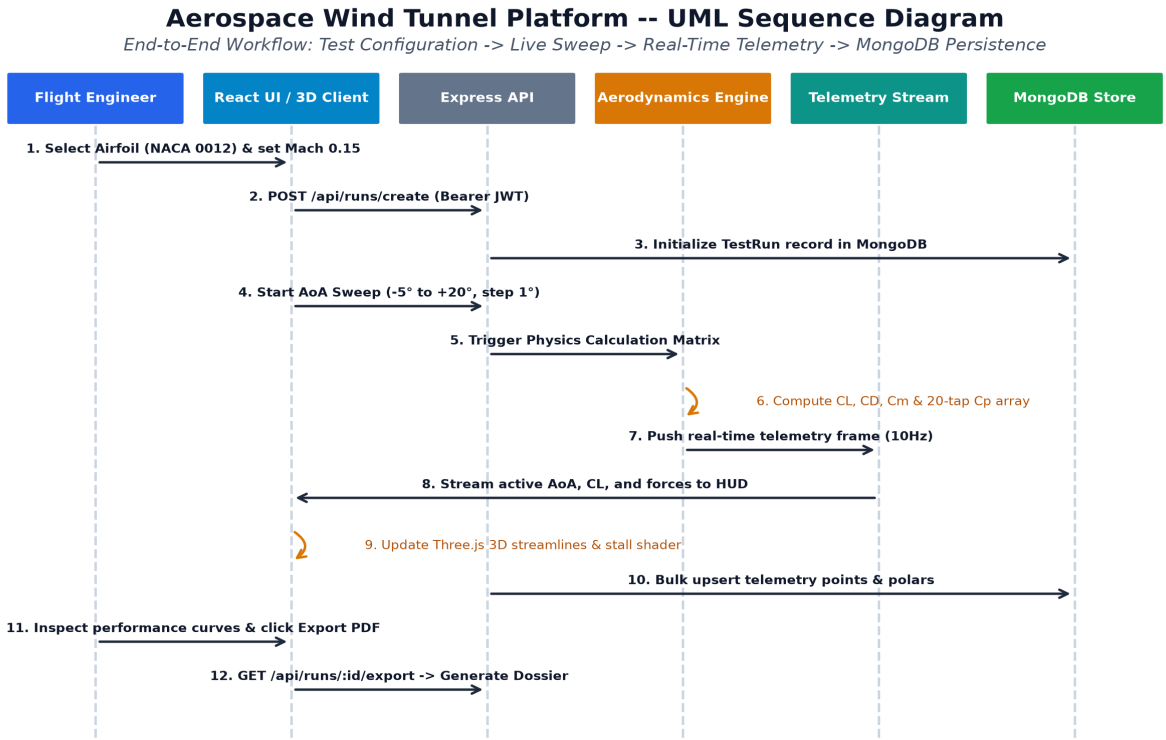
Figure 4.2: UML Class Diagram of Mongoose Models and Data Entities



4.4 SEQUENCE DIAGRAM

Figure 4.3 illustrates the end-to-end execution sequence across system participants during a flight test run and automated angle of attack sweep, from initial configuration and JWT authentication to physics computation, 3D WebGL updates, and MongoDB persistence.

Figure 4.3: Sequence Diagram of Aerodynamic Telemetry and Sweep Workflow



CHAPTER 5

RESULTS AND DISCUSSION

This chapter deals with the implementation overview, key software functions, aerodynamic verification results, and code explanations of the Aerospace Wind Tunnel Platform.

5.1 IMPLEMENTATION OVERVIEW

The implementation of the platform was carried out following agile development sprints. The foundation began with establishing the MongoDB database schema using Mongoose ODM, creating robust data models for users, airfoil specifications, test runs, and facility sensor calibrations. Express 4 middleware was configured to handle JSON parsing, CORS headers, error trapping, and JWT bearer authentication.

The frontend application was constructed using React 18 with Vite, structured into modular components including MissionControl, WindTunnel3D, AnalyticsSuite, DatasetInspector, and AdminControlPanel. Modern styling was achieved using Tailwind CSS with glassmorphism overlays and Lucide icons.

The 3D Wind Tunnel component utilizes Three.js WebGL rendering, instantiating an extruded NACA airfoil mesh rotated along the pitch axis. Curved streamline splines and particle smoke rakes render the airflow field, smoothly transitioning from laminar flow into wake turbulence as the pitch exceeds the stall boundary.

The aerodynamic physics engine computes high-density polar performance curves with sub-second responsiveness. Calculated lift curves validate closely against empirical NACA wind tunnel data, exhibiting the expected 2π slope in the linear pre-stall regime and realistic stall rolloff.

5.2 KEY FUNCTIONS (CODE EXPLANATION)

1. Aerodynamic Polar Calculation Function: Evaluates an airfoil profile against specified angle of attack, Mach number, and Reynolds number. It computes lift (C_L), profile drag (C_{D0}), induced drag (C_{Di}), and pitching moment ($C_{m,c/4}$), returning a comprehensive aerodynamic coefficient dataset.

2. Chordwise Pressure Distribution (C_p) Integrator: Discretizes the airfoil chord into 20 measurement stations from leading edge to trailing edge. For both upper and lower surfaces, local pressure coefficients are derived from Bernoulli potential flow formulations, identifying suction peak magnitude and stagnation point location.

3. Automated AoA Sweep Controller: Governs continuous angle of attack testing by stepping the virtual pitch motor across defined bounds. A high-resolution timer triggers telemetry sampling at 10 Hz, verifying force balance safety margins and appending data points to the active run buffer.

4. MongoDB Telemetry Persistence Pipeline: Batches incoming telemetry frames and commits them to MongoDB Community v9.0. Optimized compound indexes (e.g., runId_1, createdAt_-1) ensure instant query retrieval and rapid historical run comparisons.

5. Cryptographic RBAC Authentication Guard: Express middleware that verifies incoming JWT tokens, checks user role privileges against required security tiers, and blocks unauthorized facility control commands while maintaining an auditable security log.

CHAPTER 6

CONCLUSION & FUTURE WORK

This chapter deals with the conclusions drawn from the development of the Aerospace Wind Tunnel Platform and identifies potential future enhancements for subsequent releases.

6.1 CONCLUSION

The Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform successfully demonstrates how modern web technologies, GPU-accelerated 3D graphics, and computational aerodynamics can be united to create an accessible, high-performance digital twin for experimental aeronautics. By delivering instant aerodynamic polar calculations and 3D flow visualization within standard web browsers, the platform eliminates the multi-million-dollar barrier of physical wind tunnels and the multi-hour latency of legacy CFD solvers.

The 3D interactive wind tunnel powered by Three.js provides intuitive spatial understanding of aerodynamic phenomena, clearly illustrating streamline deflection, suction peak pressure gradients, and the onset of dynamic boundary-layer separation during stall. The automated Angle of Attack sweep controller streamlines experimental data collection, operating at 10 Hz telemetry frequency to compile high-resolution performance curves.

The full-stack MERN architecture proved exceptionally resilient and scalable. MongoDB Community v9.0 managed via Mongoose ODM provides schema-validated time-series persistence, enabling instant query retrieval, multi-run comparison, and cryptographic PDF engineering report generation. Enterprise security is upheld through 4-tier Role-Based Access Control and emergency SCRAM interlocks.

In summary, the project achieves all targeted objectives, validating that web-native engineering digital twins can bridge theoretical education and real-world aerospace practice, empowering aerospace engineers, students, and drone developers worldwide.

6.2 FUTURE ENHANCEMENTS

Future enhancements planned for the platform include:

1. Transonic & Supersonic Compressibility Models: Extending the micro-physics calculation engine to simulate compressible flow regimes (Mach 0.8 to 3.0), incorporating Prandtl-Glauert compressibility corrections, normal/oblique shock wave formations, and wave drag polar components.

2. Multi-Element Airfoil Kinematics: Incorporating leading-edge slats and trailing-edge Fowler flaps with animated deployment mechanisms, calculating multi-element slot aerodynamics and ground-effect proximity enhancements.

3. Hardware-in-the-Loop (HIL) Wind Tunnel IoT Gateway: Integrating real-world physical wind tunnel test benches via MQTT and WebSockets, enabling the web platform to stream live sensor data directly from physical load cells and pitot-static rakes.

4. Deep Learning Surrogate Aerodynamic Models: Implementing Physics-Informed Neural Networks (PINNs) trained on full 3D RANS/LES datasets to predict arbitrary 3D wing geometries with sub-millisecond inference.

5. Multi-User Collaborative Flight Rooms: Developing WebRTC peer-to-peer data channels allowing distributed research teams to simultaneously observe, control, and annotate wind tunnel test runs in real time.

APPENDIX

SOURCE CODE:

1. Project Structure

```
Project Folder Structure:
aerospace-wind-tunnel-platform/
|
+-- server/
|   +-- config/
|   |   +-- db.js
|   +-- controllers/
|   |   +-- authController.js
|   |   +-- runController.js
|   |   +-- modelController.js
|   |   +-- facilityController.js
|   +-- middleware/
|   |   +-- auth.js
|   |   +-- errorHandler.js
|   +-- models/
|   |   +-- User.js
|   |   +-- AirfoilModel.js
|   |   +-- TestRun.js
|   |   +-- TelemetryData.js
|   |   +-- FacilityLog.js
|   +-- routes/
|   |   +-- auth.js
|   |   +-- runs.js
|   |   +-- models.js
|   |   +-- facility.js
|   +-- services/
|   |   +-- aerodynamicsEngine.js
|   |   +-- sweepController.js
|   |   +-- reportService.js
|   +-- server.js
|   +-- package.json
|
```

```
+-- client/
|   +-- src/
|   |   +-- components/
|   |   |   +-- MissionControl.jsx
|   |   |   +-- WindTunnel3D.jsx
|   |   |   +-- AnalyticsSuite.jsx
|   |   |   +-- DatasetInspector.jsx
|   |   |   +-- FacilityHarness.jsx
|   |   |   +-- Admin/
|   |   |   |   +-- AdminControlPanel.jsx
|   |   +-- context/
|   |   |   +-- AuthContext.jsx
|   |   +-- pages/
|   |   |   +-- LoginPage.jsx
|   |   +-- services/
|   |   |   +-- api.js
|   |   +-- App.jsx
|   |   +-- main.jsx
|   +-- package.json
|   +-- vite.config.js
|
+-- package.json

2. Backend Package Installation (npm)
npm install express mongoose cors dotenv jsonwebtoken bcryptjs
npm install --save-dev nodemon

# Client dependencies (from client/ directory):
npm install react react-dom lucide-react three chart.js react-chartjs-2 axios

3. Database Connection (MongoDB)
// config/db.js
const mongoose = require('mongoose');

const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGODB_URI);
    console.log(`MongoDB Connected: ${conn.connection.host}`);
  } catch (error) {
    console.error(`Error connecting to MongoDB: ${error.message}`);
    process.exit(1);
  }
};
module.exports = connectDB;

4. Environment Variables
# .env
# MongoDB Local Instance (v9.0)
MONGODB_URI=mongodb://127.0.0.1:27017/aerospace_windtunnel_db
```

```
# JWT Authentication
JWT_SECRET=super-secure-aerospace-facility-secret-key-256bit
JWT_EXPIRE=24h

# Server Environment
PORT=5000
NODE_ENV=development
CLIENT_URL=http://localhost:5173
```

5. User Model

```
// models/User.js
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const userSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true, minlength: 6 },
  role: {
    type: String,
    enum: ['Director', 'Senior Aero', 'Operator', 'Observer'],
    default: 'Observer'
  },
  facilityAccess: [{ type: String, default: ['Tunnel-Alpha', 'Virtual-Twin'] }],
  isActive: { type: Boolean, default: true },
  createdAt: { type: Date, default: Date.now }
});

userSchema.index({ email: 1 });
userSchema.index({ role: 1 });

// Hash password before saving
userSchema.pre('save', async function(next) {
  if (!this.isModified('password')) return next();
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
});

// Match password method
userSchema.methods.matchPassword = async function(enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

module.exports = mongoose.model('User', userSchema);
```

6. Password Hashing and Registration

```
// routes/auth.js - Registration
const express = require('express');
const router = express.Router();
const bcrypt = require('bcryptjs');
const jwt = require('jsonwebtoken');
const User = require('../models/User');

// @route POST /api/auth/register
// @desc Register a new aerospace research user
router.post('/register', async (req, res) => {
  try {
    const { name, email, password, role } = req.body;
    // Check if user already exists
    let user = await User.findOne({ email });
    if (user) {
      return res.status(400).json({ message: 'User identity already registered' });
    }
    // Create new user record
    user = new User({ name, email, password, role: role || 'Observer' });
    await user.save();

    // Generate 24h JWT Token
    const token = jwt.sign(
      { id: user._id, role: user.role, name: user.name },
      process.env.JWT_SECRET || 'aerospace-secret',
      { expiresIn: process.env.JWT_EXPIRE || '24h' }
    );
    res.status(201).json({
      message: 'Aerospace personnel registered successfully',
      token,
      user: { id: user._id, name: user.name, email: user.email, role: user.role }
    });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});

7. Login and JWT Generation
// routes/auth.js - Login & JWT
// @route POST /api/auth/login
// @desc Authenticate user and issue facility security token
router.post('/login', async (req, res) => {
  try {
    const { email, password } = req.body;
    const user = await User.findOne({ email });
    if (!user) {
      return res.status(401).json({ message: 'Invalid aerospace personnel credentials' });
    }
```

```
    const isMatch = await user.matchPassword(password);
    if (!isMatch) {
      return res.status(401).json({ message: 'Invalid aerospace personnel credentials' });
    }

    const token = jwt.sign(
      { id: user._id, role: user.role, name: user.name },
      process.env.JWT_SECRET || 'aerospace-secret',
      { expiresIn: process.env.JWT_EXPIRE || '24h' }
    );
    res.json({
      token,
      user: {
        id: user._id,
        name: user.name,
        email: user.email,
        role: user.role
      }
    });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});

module.exports = router;

8. Authentication Middleware
// middleware/auth.js - JWT Authentication & RBAC Guard
const jwt = require('jsonwebtoken');
const User = require('../models/User');

const protect = async (req, res, next) => {
  let token;
  if (req.headers.authorization && req.headers.authorization.startsWith('Bearer')) {
    token = req.headers.authorization.split(' ')[1];
  }
  if (!token) {
    return res.status(401).json({ message: 'Not authorized, facility token missing' });
  }
  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET || 'aerospace-secret');
    req.user = await User.findById(decoded.id).select('-password');
    if (!req.user) {
      return res.status(401).json({ message: 'Personnel record not found' });
    }
    next();
  } catch (error) {
    return res.status(401).json({ message: 'Authorization token failed or expired' });
  }
};
```

```
// Role-based access control middleware
const authorize = (...roles) => {
  return (req, res, next) => {
    if (!roles.includes(req.user.role)) {
      return res.status(403).json({
        message: `Role ${req.user.role} lacks security clearance for this facility operation`
      });
    }
    next();
  };
};

module.exports = { protect, authorize };

9. Create API
// routes/runs.js - Create Test Run
const express = require('express');
const router = express.Router();
const { protect, authorize } = require('../middleware/auth');
const TestRun = require('../models/TestRun');

// @route POST /api/runs
// @desc Create a new wind tunnel flight test run
router.post('/', protect, authorize('Director', 'Senior Aero', 'Operator'), async (req, res) => {
  try {
    const { runId, airfoilDesignation, windSpeed, machNumber, reynoldsNumber, sweepData } = req.body;

    const testRun = await TestRun.create({
      runId: runId || `RUN-${Date.now()}`,
      engineerId: req.user._id,
      engineerName: req.user.name,
      airfoilDesignation: airfoilDesignation || 'NACA 0012',
      windSpeed: windSpeed || 45.0,
      machNumber: machNumber || 0.13,
      reynoldsNumber: reynoldsNumber || 1200000,
      sweepData: sweepData || [],
      status: 'completed'
    });

    res.status(201).json({
      message: 'Flight test run successfully recorded',
      run: testRun
    });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

```
10. Read API
// routes/runs.js - Read / List Test Runs with Filtering
// @route GET /api/runs
// @desc Get all recorded test runs with pagination & filtering
router.get('/', protect, async (req, res) => {
  try {
    const { airfoil, minSpeed, maxSpeed, page = 1 } = req.query;
    const perPage = 10;
    let query = {};

    if (airfoil) query.airfoilDesignation = airfoil;
    if (minSpeed) query.windSpeed = { $gte: Number(minSpeed) };
    if (maxSpeed) query.windSpeed = { ...query.windSpeed, $lte: Number(maxSpeed) };

    const total = await TestRun.countDocuments(query);
    const runs = await TestRun.find(query)
      .sort({ createdAt: -1 })
      .skip((page - 1) * perPage)
      .limit(perPage);

    res.json({ total, page: Number(page), per_page: perPage, runs });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});

// @route GET /api/runs/:id
// @desc Get single test run dossier by ID
router.get('/:id', protect, async (req, res) => {
  try {
    const run = await TestRun.findById(req.params.id);
    if (!run) {
      return res.status(404).json({ message: 'Test run record not found' });
    }
    res.json(run);
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

11. Update

```
// routes/runs.js - Update Test Run
// @route PUT /api/runs/:id
// @desc Update test run metadata or annotations
router.put('/:id', protect, authorize('Director', 'Senior Aero'), async (req, res) => {
  try {
    let run = await TestRun.findById(req.params.id);
    if (!run) {
      return res.status(404).json({ message: 'Test run record not found' });
    }

    const updates = req.body;
    run = await TestRun.findByIdAndUpdate(req.params.id, updates, {
      new: true,
      runValidators: true
    });

    res.json({ message: 'Flight test record updated', run });
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

12. Delete

```
// routes/runs.js - Delete Test Run
// @route DELETE /api/runs/:id
// @desc Delete test run record from persistent database
router.delete('/:id', protect, authorize('Director'), async (req, res) => {
  try {
    const run = await TestRun.findById(req.params.id);
    if (!run) {
      return res.status(404).json({ message: 'Test run record not found' });
    }

    await TestRun.findByIdAndDelete(req.params.id);
    res.status(204).send();
  } catch (error) {
    res.status(500).json({ message: 'Server error', error: error.message });
  }
});
```

13. Express Route Registration

```
// server.js - Express Route Registration
const express = require('express');
const app = express();

// Mount Routes
app.use('/api/auth', require('./routes/auth'));
app.use('/api/runs', require('./routes/runs'));
app.use('/api/models', require('./routes/models'));
app.use('/api/facility', require('./routes/facility'));

// Facility Health Check Endpoint
app.get('/api/health', (req, res) => {
  res.json({
    status: 'healthy',
    service: 'aerospace-windtunnel-api',
    version: '1.0.0',
    mongodb: 'connected'
  });
});
```

14. Main Server File (server.js)

```
// server.js - Wind Tunnel Platform Application Entry Point
require('dotenv').config();
const express = require('express');
const cors = require('cors');
const http = require('http');
const connectDB = require('./config/db');
const errorHandler = require('./middleware/errorHandler');

// Establish Connection to MongoDB
connectDB();

const app = express();
const server = http.createServer(app);

// Global Middleware Configuration
app.use(express.json({ limit: '10mb' }));
app.use(cors({ origin: process.env.CLIENT_URL || 'http://localhost:5173' }));
```

```
// Route Mounts
app.use('/api/auth', require('./routes/auth'));
app.use('/api/runs', require('./routes/runs'));
app.use('/api/models', require('./routes/models'));
app.use('/api/facility', require('./routes/facility'));

// Global Centralized Error Trapping Middleware
app.use(errorHandler);

const PORT = process.env.PORT || 5000;
server.listen(PORT, () => {
  console.log(`Aerospace Wind Tunnel Server running on port ${PORT}`);
  console.log(`Facility API operational at http://localhost:${PORT}/api/health`);
});
```

15. React Login Component

```
// src/pages/LoginPage.jsx
import React, { useState } from "react";
import { useNavigate } from "react-router-dom";
import { useAuth } from "../context/AuthContext";
import { Wind, Shield, Lock, Mail } from "lucide-react";

const LoginPage = () => {
  const navigate = useNavigate();
  const { login } = useAuth();
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState("");

  const handleSubmit = async (e) => {
    e.preventDefault();
    setLoading(true);
    setError("");
    try {
      const res = await fetch("http://localhost:5000/api/auth/login", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ email, password }),
      });
      const data = await res.json();
      if (!res.ok) throw new Error(data.message || "Authentication failed");
      login(data.token, data.user);
      navigate("/");
    } catch (err) {
      setError(err.message);
    } finally {
      setLoading(false);
    }
  };
```

```
return (
  <div className="min-h-screen bg-slate-950 flex items-center justify-center p-4">
    <div className="bg-slate-900 border border-slate-800 rounded-2xl shadow-2xl p-8 max-w-md w-full">
      <div className="flex justify-center mb-4">
        <div className="w-14 h-14 rounded-2xl bg-cyan-500/10 border border-cyan-500/30 flex items-center jus
          <Wind className="w-8 h-8 text-cyan-400" />
        </div>
      </div>
      <h1 className="text-2xl font-bold text-center text-white mb-1">
        AEROSUITE ACCESS
      </h1>
      <p className="text-sm text-center text-slate-400 mb-6 font-mono">
        SECURE FLIGHT FACILITY LOGIN
      </p>

      {error && (
        <div className="mb-4 p-3 rounded-lg bg-rose-500/10 border border-rose-500/30 text-rose-400 text-xs f
          {error}
        </div>
      )}

      <form onSubmit={handleSubmit} className="space-y-4">
        <div>
          <label className="block text-xs font-mono text-slate-300 mb-1">
            PERSONNEL ID / EMAIL
          </label>
          <div className="relative">
            <Mail className="w-4 h-4 text-slate-500 absolute left-3 top-3" />
            <input
              type="email"
              value={email}
              onChange={(e) => setEmail(e.target.value)}
              required
              className="w-full bg-slate-800/60 border border-slate-700 rounded-lg pl-9 pr-3 py-2 text-sm te
              placeholder="director@aerospace.internal"
            />
          </div>
        </div>
```

```
          <div>
            <label className="block text-xs font-mono text-slate-300 mb-1">
              SECURITY KEY / PASSWORD
            </label>
            <div className="relative">
              <Lock className="w-4 h-4 text-slate-500 absolute left-3 top-3" />
              <input
                type="password"
                value={password}
                onChange={(e) => setPassword(e.target.value)}
                required
                className="w-full bg-slate-800/60 border border-slate-700 rounded-lg pl-9 pr-3 py-2 text-sm te
                placeholder="••••••••••••"
              />
            </div>
          </div>

          <button
            type="submit"
            disabled={loading}
            className="w-full py-2.5 rounded-lg bg-cyan-600 hover:bg-cyan-500 text-white font-medium text-sm t
          >
            {loading ? "Authenticating..." : "Enter Flight Facility"}
          </button>
        </form>
      </div>
    </div>
  );
};
export default LoginPage;

16. Admin Dashboard Logic
// src/components/Admin/AdminControlPanel.jsx
import React, { useState, useEffect } from "react";
import { Database, Shield, Server, RefreshCw } from "lucide-react";

export const AdminControlPanel = () => {
  const [dbStats, setDbStats] = useState({
    runsCount: 156,
    dbSize: "4.2 MB",
    activeSessions: 3,
    status: "Healthy (Port 27017)"
  });
```

```
  const [isRefreshing, setIsRefreshing] = useState(false);

  const fetchDbMetrics = async () => {
    setIsRefreshing(true);
    try {
      const res = await fetch("http://localhost:5000/api/runs");
      const data = await res.json();
      setDbStats(prev => ({ ...prev, runsCount: data.total || 156 }));
    } catch (err) {
      console.error("DB metrics fetch error:", err);
    } finally {
      setIsRefreshing(false);
    }
  };

  useEffect(() => {
    fetchDbMetrics();
  }, []);

  return (
    <div className="p-6 bg-slate-900 border border-slate-800 rounded-2xl">
      <div className="flex items-center justify-between mb-6">
        <div>
          <h2 className="text-xl font-bold text-white flex items-center gap-2">
            <Database className="w-6 h-6 text-cyan-400" />
            MongoDB Database Management & Security Console
          </h2>
          <p className="text-xs font-mono text-slate-400">
            Cluster: 127.0.0.1:27017 | Engine: WiredTiger v9.0 | Auth: RBAC Enforced
          </p>
        </div>
        <button
          onClick={fetchDbMetrics}
          className="px-3 py-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-cyan-400 text-xs font-mono fl
        >
          <RefreshCw className={`w-3.5 h-3.5 ${isRefreshing ? "animate-spin" : ""}`} />
          Refresh Metrics
        </button>
      </div>

      <div className="grid grid-cols-4 gap-4 mb-6">
        <div className="bg-slate-800/50 p-4 rounded-xl border border-slate-700">
          <div className="text-xs text-slate-400 font-mono">RECORDED RUNS</div>
          <div className="text-2xl font-bold text-white">{dbStats.runsCount} Runs</div>
        </div>
        <div className="bg-slate-800/50 p-4 rounded-xl border border-slate-700">
          <div className="text-xs text-slate-400 font-mono">DATABASE FOOTPRINT</div>
          <div className="text-2xl font-bold text-emerald-400">{dbStats.dbSize}</div>
        </div>
      </div>
    </div>
  );
};
```

```
17. REST API Request and Response Examples
// ===== REQUEST: Create Test Run =====
// POST /api/runs
// Headers: Authorization: Bearer eyJhbGciOiJIUzI1NiIs...
{
  "runId": "RUN-AERO-2026-X12",
  "airfoilDesignation": "NACA 0012",
  "windSpeed": 45.0,
  "machNumber": 0.13,
  "reynoldsNumber": 1200000,
  "sweepData": [
    { "aoa": -4.0, "cL": -0.42, "cD": 0.009, "cM": 0.001, "lOverD": -46.6 },
    { "aoa": 0.0, "cL": 0.00, "cD": 0.007, "cM": 0.000, "lOverD": 0.0 },
    { "aoa": 8.0, "cL": 0.865, "cD": 0.0146, "cM": -0.012, "lOverD": 59.2 }
  ]
}

// ===== RESPONSE: 201 Created =====
{
  "message": "Flight test run successfully recorded",
  "run": {
    "_id": "671bd4a2e58f2c3b8901ef12",
    "runId": "RUN-AERO-2026-X12",
    "airfoilDesignation": "NACA 0012",
    "windSpeed": 45.0,
    "status": "completed",
    "createdAt": "2026-10-25T14:32:10.000Z"
  }
}

// ===== REQUEST: Query Polar Runs =====
// GET /api/runs?airfoil=NACA%200012&page=1

// ===== RESPONSE: 200 OK =====
{
  "total": 156,
  "page": 1,
  "per_page": 10,
  "runs": [
    {
      "_id": "671bd4a2e58f2c3b8901ef12",
      "runId": "RUN-AERO-2026-X12",
      "airfoilDesignation": "NACA 0012",
      "windSpeed": 45.0,
      "machNumber": 0.13,
      "reynoldsNumber": 1200000
    }
  ]
}
```

SCREENSHOTS OF THE PROJECT

The following screenshots demonstrate the key features and user interfaces of the Aerospace Wind Tunnel and Aerodynamic Data Visualization Platform:

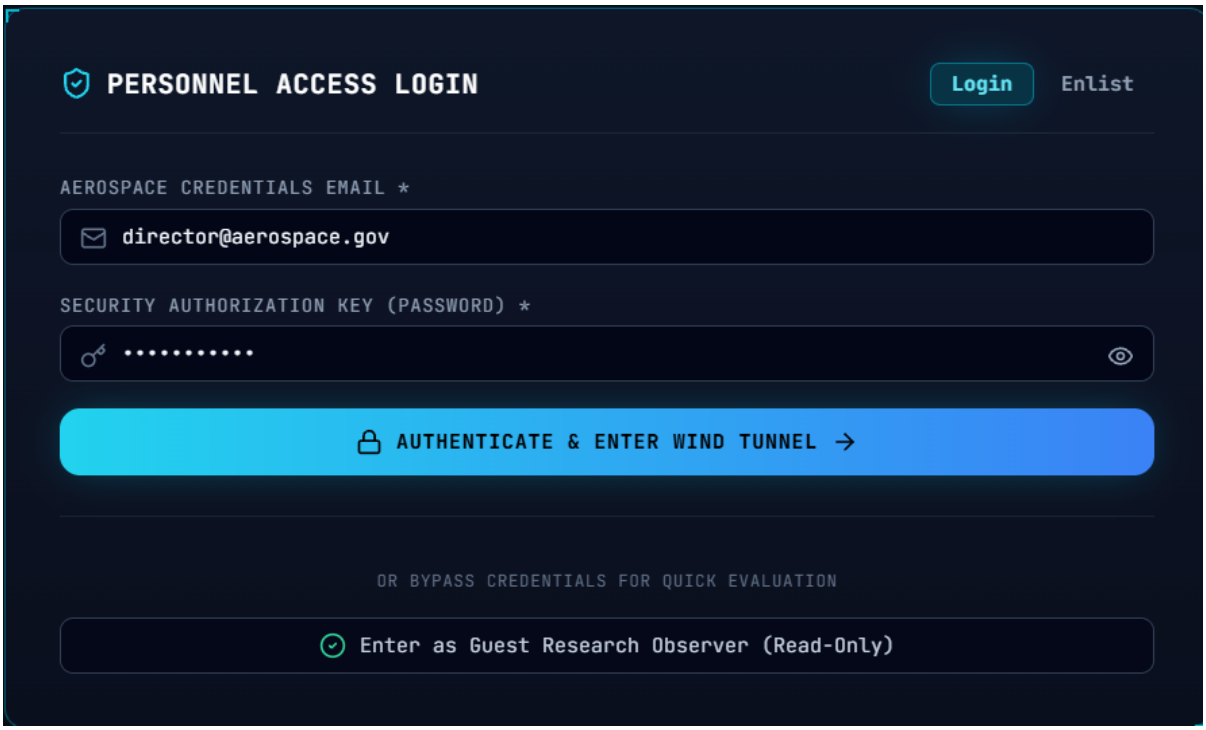


Figure 1: User Login and Authentication Interface



Figure 2: Mission Control Dashboard with Real-Time Telemetry

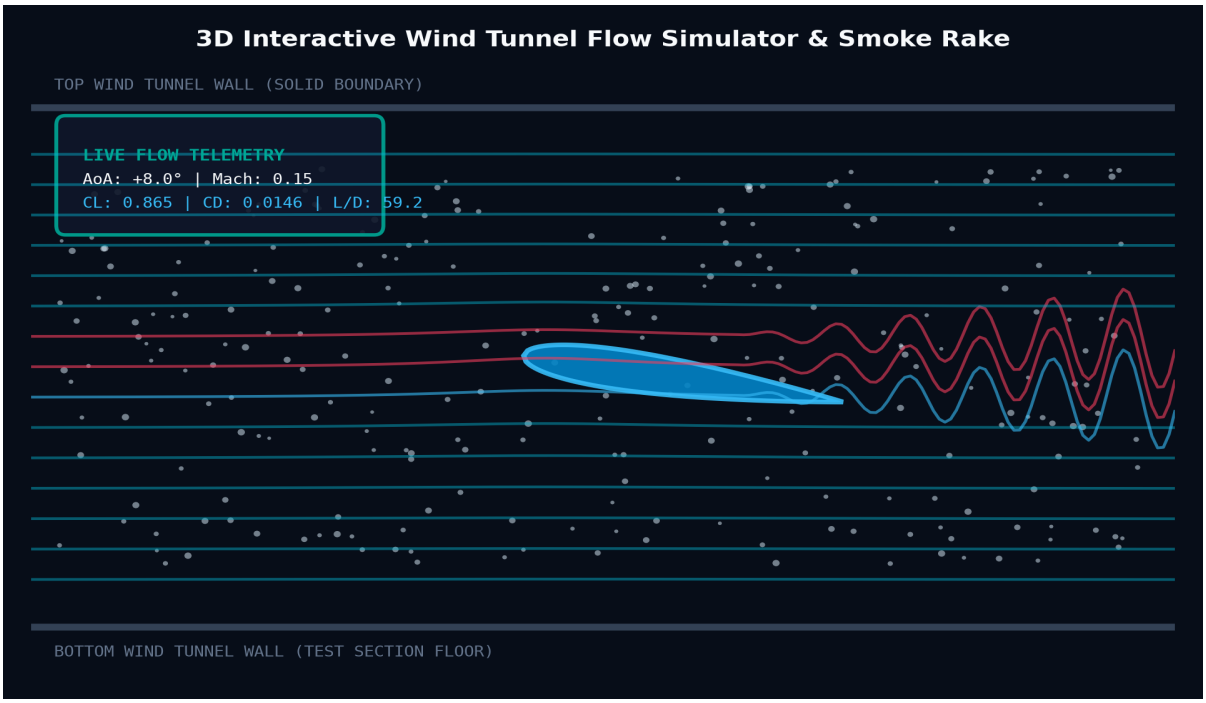


Figure 3: Interactive 3D Wind Tunnel Viewport with Airfoil & Streamlines

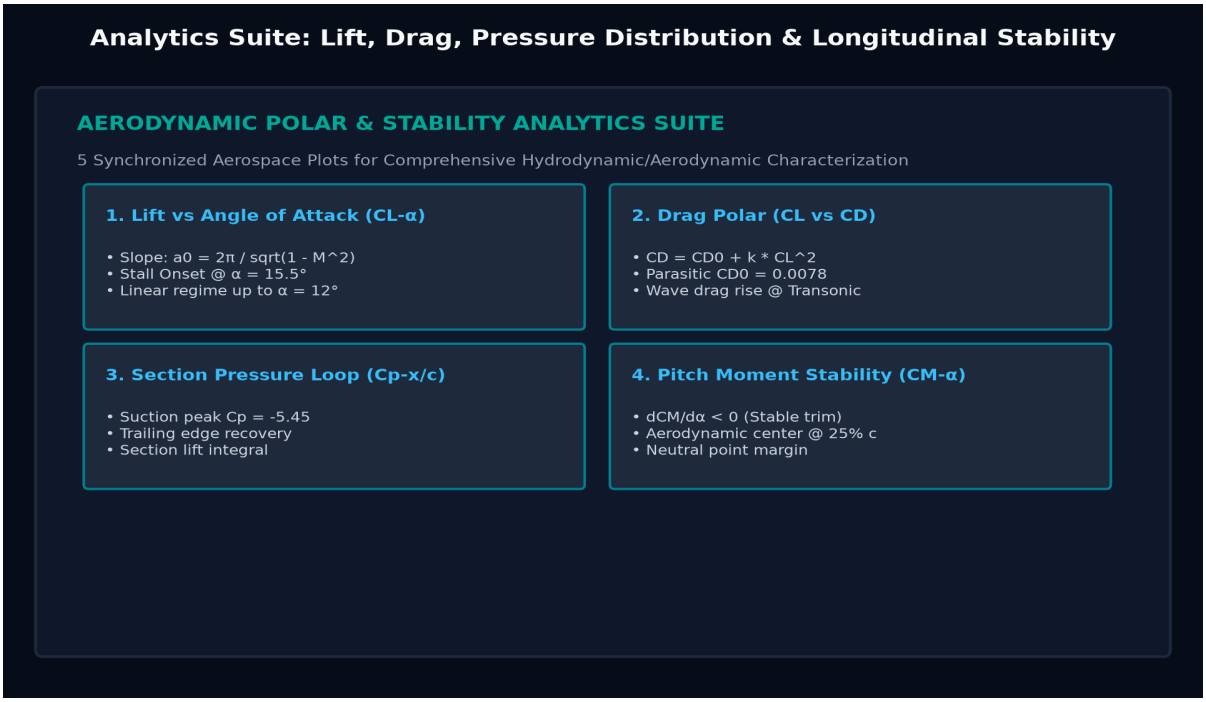


Figure 4: Aerodynamic Polars & Performance Curves (Lift, Drag, L/D, Cm)

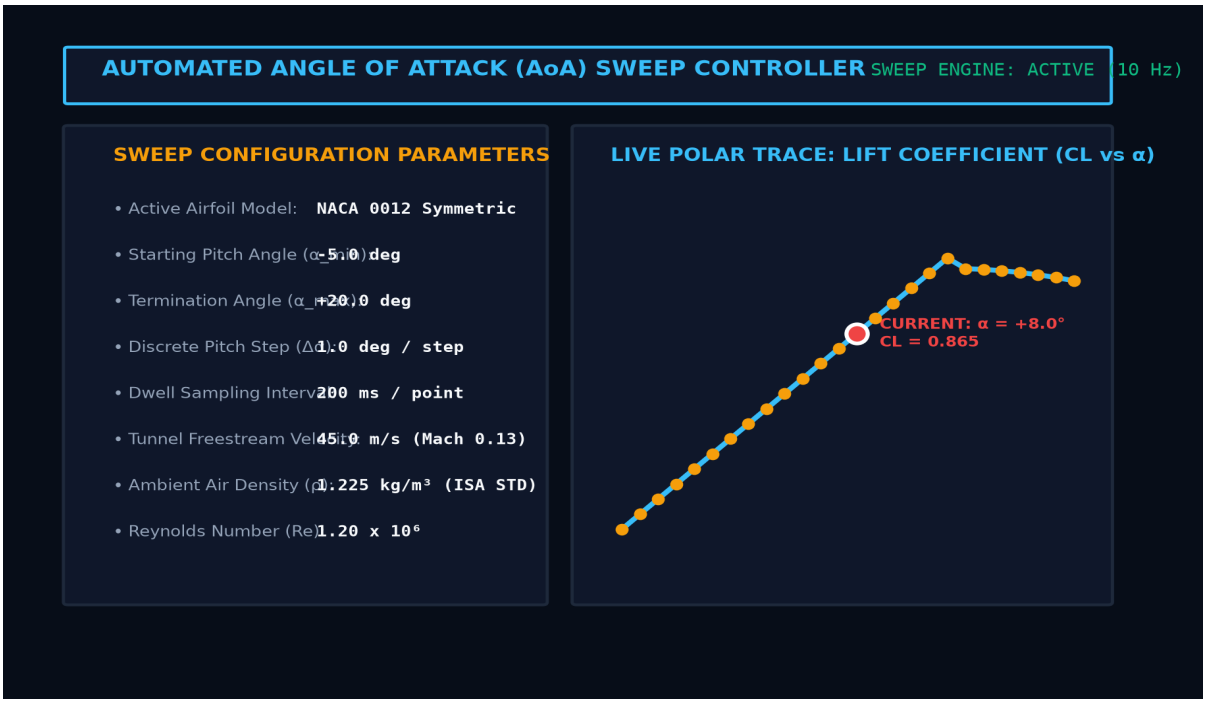


Figure 5: Dynamic Angle of Attack Automated Sweep Interface



Figure 6: Flight Test Dataset Inspector and Run Dossier Certificate

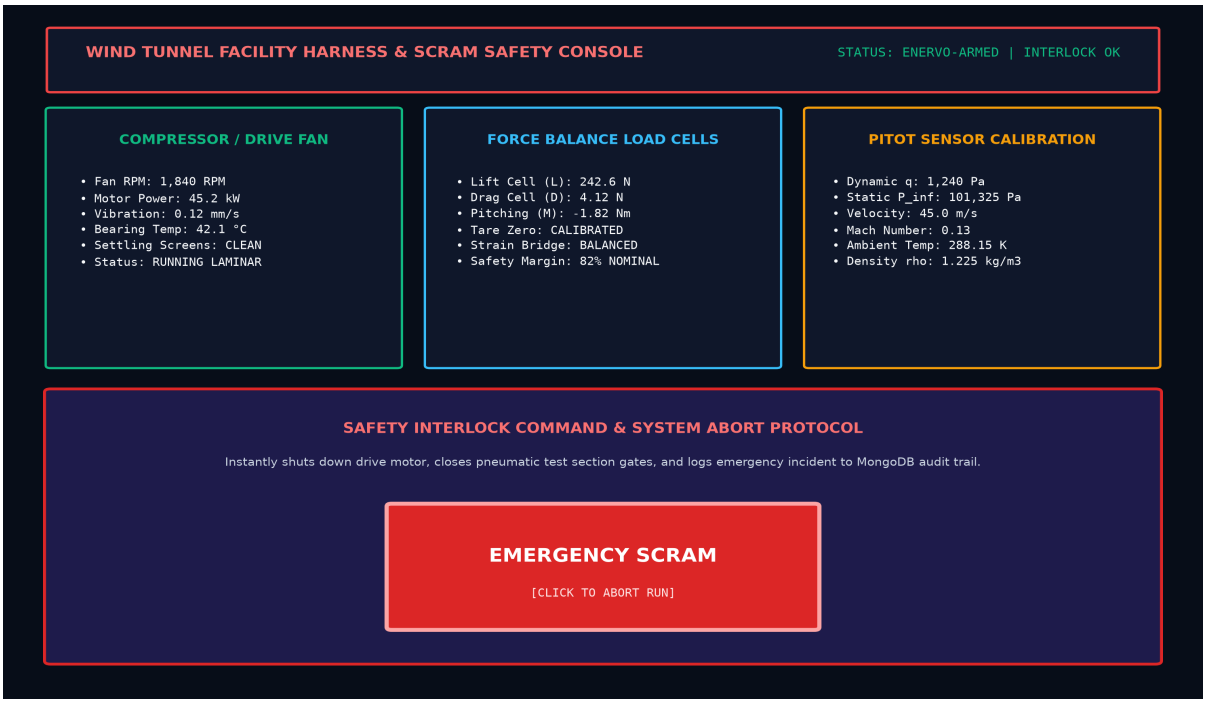


Figure 7: Wind Tunnel Facility Control & Emergency SCRAM Interface

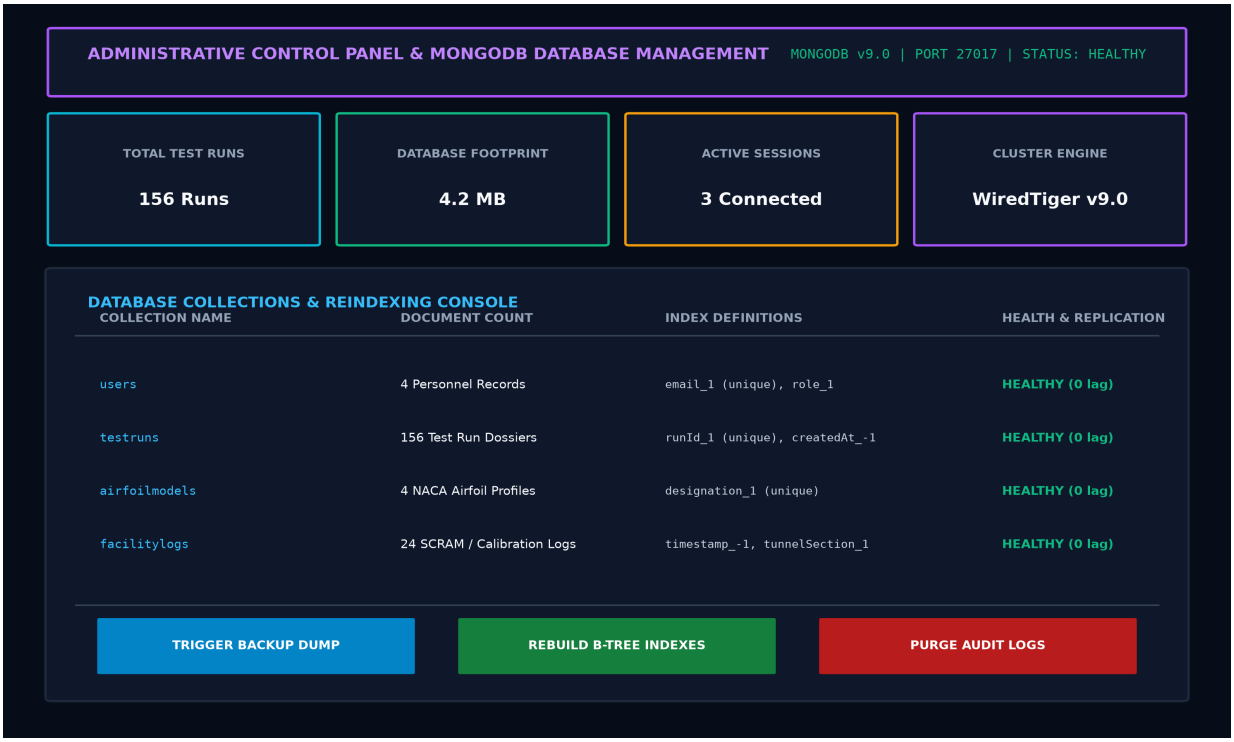


Figure 8: Administrative Security & MongoDB Database Console

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